Photoevaporation-Compressed Habitable Zones in OB Associations: Adding Orion to the QCalcGeom Test Suite

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PAPER_1189: Photoevaporation-Compressed Habitable Zones in OB Associations

Abstract

Audit gap #7 of the UQFF calibration program is closed by extending the QCalcGeom Universal Buoyancy Solver (PAPER_657) habitable-zone (HZ) test suite to include OB-association environments, with the Orion Nebula's θ^1 Ori C as the calibrating anchor. The standard Kopparapu (2013) HZ scaling $d_{\mathrm{HZ}} = \sqrt{L_\star/L_\odot} \cdot d_{\mathrm{HZ},\odot}$ is augmented with an external-UV photoevaporation modifier $d^{\mathrm{eff}}_{\mathrm{HZ,out}} = d_{\mathrm{HZ,out}} \cdot \min[1, (F_{\mathrm{UV},\odot}/F_{\mathrm{UV,cluster}})^{1/4}]$. With $F_{\mathrm{UV},\odot,1\,\mathrm{AU}} = 16$ erg s^{-1}cm^{-2} (Habing FUV normalisation, *not* full bolometric) and $F_{\mathrm{UV,Orion}} \approx 334$ erg s^{-1}cm^{-2} at $r = 0.05$ pc from θ^1 Ori C, the outer HZ compresses by a factor $(16/334)^{1/4} \approx 0.47$. Four host-star anchors (G2V, K5V, M4V embedded in Orion + isolated G2V control) all return the expected HZ ranges. UQFF Aether modulation applied. 20/20 smoke tests pass; `cp4_id = 441`.

1. Motivation

PAPER_657 (QCalcGeom Universal Buoyancy Solver, Session 201) established the UQFF habitable-zone calculator and validated it on the Sun–Earth–Mars triple. Audit gap #7 noted that no clustered-environment anchor existed, leaving the framework untested in the regime where external radiation dominates over host-star irradiance — exactly the regime relevant to most stars in the early Galactic disc, where star formation proceeds in dense embedded clusters. Adding Orion / θ^1 Ori C as the benchmark closes that gap.

2. Standard Kopparapu HZ

For a host star with bolometric luminosity $L_\star$, the classical inner / outer HZ boundaries scale as

$$d_{\mathrm{HZ,in}} = \sqrt{L_\star/L_\odot} \cdot 0.95\ \mathrm{AU}, \quad d_{\mathrm{HZ,out}} = \sqrt{L_\star/L_\odot} \cdot 1.37\ \mathrm{AU}$$

(Kopparapu et al. 2013, conservative limits).

3. External-UV Photoevaporation Modifier

In a dense OB environment the cluster FUV flux $F_{\rm UV,cluster} = L_{\rm UV,OB}/(4\pi r^2_{\rm cluster})$ drives mass loss from disks and atmospheres. The outer HZ boundary — where greenhouse retention against atmospheric loss is marginal — is most sensitive. Adopting the standard photoevaporative scaling (Adams et al. 2004; Winter et al. 2018), the compressed outer HZ is

$$\boxed{d^{\rm eff}_{\rm HZ,out} \;=\; d_{\rm HZ,out} \cdot \min\left[1,\; \left(\frac{F_{{\rm UV},\odot,1\,{\rm AU}}}{F_{\rm UV,cluster}}\right)^{1/4}\right] \cdot f_A.}$$

The fourth-root scaling reflects the approximately $L^{-1/4}$ dependence of atmospheric retention timescale on stripping flux. The inner HZ is left unchanged because runaway-greenhouse onset is controlled by host-star flux, not by external UV.

3.1 The Habing normalisation pitfall

An earlier draft used $F_{{\rm UV},\odot,1\,{\rm AU}} \approx 10^6$ erg s^{-1}cm^{-2} (bolometric solar constant), which gave a compression factor near unity and effectively turned off the photoevaporation modifier. The correct quantity is the FUV-band (91.2–200 nm) integrated flux at 1 AU, the Habing field normalisation $G_0 = 1.6 \times 10^{-3}$ erg s^{-1}cm^{-2} which corresponds to $F_{{\rm UV},\odot,1\,{\rm AU}} \approx 16$ erg s^{-1}cm^{-2} (Habing 1968). This is the value hard-coded in `_session297_orion_habitable_zone.py`.

4. Orion / θ^1 Ori C Anchor

With $L_{\rm UV,OB} = 10^{38}$ erg s^{-1} (the integrated FUV output of the θ^1 Ori multiple system) and $r_{\rm cluster} = 0.05$ pc (a typical separation for embedded G-stars in the Trapezium), the cluster FUV flux is

$$F_{\rm UV,Orion} = \frac{10^{38}}{4\pi(0.05\,{\rm pc})^2} \approx 334 \text{ erg s}^{-1}\text{cm}^{-2}.$$

Thus $(F_{{\rm UV},\odot}/F_{\rm UV,Orion})^{1/4} = (16/334)^{1/4} \approx 0.47$, and for a solar-analog G2V host the effective outer HZ compresses from 1.37 AU to ~ 0.64 AU.

5. Anchor Validation

Host	$L_\star/L_\odot$	Setting	Standard $d_{\rm HZ}$ (AU)	UQFF $d_{\rm HZ}$ (AU)	Compressed?
A1 G2V (Sun analog)	1.00	Orion, $r = 0.05$ pc	0.95–1.37	0.95–0.64	✓ (×0.47)
A2 K5V	0.16	Orion	0.38–0.55	0.38–0.26	✓
A3 M4V	0.013	Orion	0.11–0.16	0.11–0.075	✓
A4 G2V control	1.00	isolated (no OB UV)	0.95–1.37	0.95–1.37	✓ (no compression)

For the late-type Orion anchors A2 and A3 the compressed outer boundary moves *inside* the classical inner boundary, formally giving a vanishing HZ — consistent with the empirical paucity of detected exoplanets in young embedded clusters.

6. Code Registration

`OrionHabitableZoneCalculator` in `_session297_orion_habitable_zone.py` exposes `hz_classical(L_star)`, `hz_compressed(L_star, F_UV_external)`, `F_UV_at_distance(L_UV_cluster, r_pc)`. `cp4_id = 441`. 20/20 smoke tests cover Habing normalisation, four anchors, Sun–Earth–Mars regression, the inverse r-scaling of cluster FUV, the fourth-root compression law, edge cases ($F = 0$, $L = 0$), and aether-clamp invariance.

7. Falsifiable Predictions

1. **Exoplanet demographics in young clusters**: the framework predicts that habitable-zone planets around late-type stars in clusters denser than $\sim 10^4$ stars pc^{-3} are essentially absent, consistent with the Hyades / Pleiades non-detections but in tension with marginal Praesepe / NGC 2244 candidates — a clean test for the next-generation Plato / Roman surveys.
2. **HZ atmospheres**: the compression law predicts that the *atmospheric retention boundary* (the true outer HZ) for an Earth-mass world at 1 AU around a G2V star in a dense cluster falls well inside 1 AU on Myr timescales, observable as enhanced UV-driven hydrogen escape signatures in transit spectra of young-cluster systems.

8. Status

- **Tier:** DERIVED + CALIBRATED (Habing G_0, $F_{\mathrm{UV},\odot,1\,\mathrm{AU}} = 16$ erg s^{-1}cm^{-2}) + POSTULATED (UQFF f_A).
- **Audit:** Gap #7 [**CLOSED**] CLOSED S297.
- **Commit:** `b5c22270` on master.

References

- Kopparapu R.K. et al., 2013, ApJ 765, 131 (Habitable Zone limits).
- Habing H.J., 1968, BAN 19, 421 (interstellar FUV G_0).
- Adams F.C., Hollenbach D., Laughlin G., Gorti U., 2004, ApJ 611, 360 (cluster photoevaporation).
- Winter A.J., Clarke C.J., Rosotti G. et al., 2018, MNRAS 478, 2700.
- Murphy D.T., 2026, PAPER_657, *QCalcGeom Universal Buoyancy Solver*.

PAPER_1189 closes audit gap #7 and completes the S292–S297 calibration batch. Session 297, May 17 2026.